PRASANTH MAMIDISETTI

Analytics / Agentic AI | GenAI | Data & Machine Learning

+1 469-839-5534 | mamidisettiprasanth36@gmail.com | LinkedIn: linkedin.com/in/mamidisetti-prasanth-b1aa92225

PROFESSIONAL SUMMARY

Generative AI and AI/ML Engineer with hands-on experience in Python, machine learning, data analysis, AI agents, model evaluation, classification, product intelligence, data preparation, and automation. Experienced analyzing and organizing structured and unstructured data, supporting product search, ranking, recommendations, benchmarking, and consumer insights, and translating analytical findings into actionable business insights. Strong interest in forecasting, emerging AI trends, automation, product innovation, and data-driven analysis of cost, quality, and product performance. Strong problem-solving, critical-thinking, communication, cross-functional collaboration, organization, project coordination, and time-management skills.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Bash, JavaScript (Basic), TypeScript (Basic), Data Preparation, Data Cleaning, Data Transformation, Data Analysis, ETL, Data Validation, Data Quality, EDA

AI / Agentic AI: Generative AI, LLMs, AI Agents, Agentic Workflows, RAG, Prompt Engineering, Model Evaluation, AI Model Testing, Classification, Anomaly Detection, Embeddings, Fine-Tuning (LoRA, QLoRA, PEFT), Function Calling, Workflow Automation

ML & Analytics: scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, Feature Engineering, Hyperparameter Tuning, Cross-Validation, A/B Testing, Forecasting, Benchmarking, Statistical Analysis, Trend Analysis, Data-Driven Insights

Product & Business Analytics: Product Analytics, Product Intelligence, Product Benchmarking, Product Performance Analysis, Product Quality Analysis, Recommendation Analysis, Consumer Insights, Market Trend Analysis, Cost Analysis Concepts

AI Innovation & Research: Emerging AI Trends, Emerging Technology Research, AI Automation, Product Innovation, AI-Assisted Analysis, Emerging Technology Evaluation

Frameworks: LangChain, LangGraph, FastAPI, spaCy, Keras

Data & Cloud: PostgreSQL, MySQL, Redis, Apache Kafka, Pinecone, FAISS, Chroma, Weaviate, Elasticsearch, AWS (S3, ECS, Lambda, SageMaker, CloudWatch), GCP (Vertex AI, BigQuery), Azure ML

Software / Productivity: Git, GitHub Actions, CI/CD, Docker, MLflow, JIRA, Postman, REST APIs, Unit Testing, Documentation, Microsoft Office (Excel, Word, PowerPoint, Outlook)

Professional: Critical Thinking, Problem Solving, Analytical Mindset, Cross-Functional Collaboration, Written & Verbal Communication, Organization, Time Management, Project Coordination, Action-Item Tracking, Attention to Detail, Proactive Learning, Teamwork

PROFESSIONAL EXPERIENCE

BTG Holdings — Software Engineer Intern | Aug 2026 – Present

• Contribute to AI-driven product intelligence initiatives supporting product data organization, product capture, product matching, search, ranking, and recommendation workflows.

• Analyze structured product data, product attributes, product reviews, and merchant-sourced information to support product discovery and consumer-focused insights.

• Support product benchmarking and comparative analysis by organizing product attributes and evaluating similarities, rankings, recommendation signals, and product data quality.

• Analyze product-preference and buying-habit signals to support consumer insights and personalized product recommendations.

• Support product performance analysis by evaluating search, ranking, matching, and recommendation outputs and identifying opportunities for improvement.

• Assist with data-quality validation across product records to improve accuracy, consistency, and reliability of product information.

• Contribute to AI/ML and automation workflows and research emerging AI and automation approaches relevant to product intelligence and consumer-facing applications.

• Map business and technical workflows to understand data flow, system dependencies, requirements, and integration points.

• Track project action items, technical tasks, testing activities, and follow-ups across ongoing development work.

• Perform API testing and validation to verify data flow, application behavior, and integration functionality.

• Collaborate with engineering and cross-functional team members to gather requirements, troubleshoot issues, validate outputs, and support product development using AI/ML, APIs, MongoDB, and related technologies.

Texas Review Pvt Ltd (EdTech AI Platform) — AI Engineer | Jan 2023 – Dec 2025

• Architected and deployed production-grade Generative AI applications using GPT-4, LLaMA 2, and Mistral for enterprise knowledge management and intelligent document processing.

• Designed RAG pipelines using LangChain, LangGraph, Pinecone, and FAISS; developed multi-agent workflows for task planning, tool invocation, and automation.

• Engineered NLP pipelines for document classification, sentiment analysis, NER, and summarization across large-scale datasets.

• Implemented document ingestion, chunking, metadata enrichment, embeddings, and vector indexing across 50,000+ enterprise documents.

• Improved classification F1-score from 0.74 to 0.86 through feature engineering, hyperparameter tuning, cross-validation, and A/B testing.

• Optimized AI inference through prompt optimization, caching, quantization, and efficient model serving, reducing latency and infrastructure costs.

• Collaborated with product managers, data scientists, and software engineers to deploy AI microservices on AWS.

• Performed testing, debugging, performance optimization, and model evaluation to improve production reliability.

• Gathered business requirements and translated AI use cases into scalable technical solutions.

Infosys Limited — System Engineer Trainee | Jul 2022 – Dec 2022

• Integrated NLP models into enterprise workflows for automated ticket classification and priority routing, reducing manual triage time by 25% across 10,000+ monthly tickets.

• Developed Python data-preprocessing pipelines to clean, transform, validate, and organize structured and unstructured datasets.

• Built anomaly-detection pipelines using Python and scikit-learn to identify data inconsistencies and improve data quality.

• Designed automated data-validation workflows to improve data accuracy and reduce manual verification.

• Optimized SQL queries and ETL workflows to improve data retrieval and support large-scale analytics.

• Performed EDA to identify trends, detect anomalies, and generate actionable insights for business stakeholders.

• Automated repetitive operational tasks using Python, saving over 2,000 hours of manual effort annually.

• Collaborated with cross-functional teams, participated in Agile/Scrum ceremonies, and documented technical designs.

CERTIFICATION

Oracle Agentic AI Certified Foundations Associate — Oracle

EDUCATION

Indiana Wesleyan University, USA — Master of Science (M.S.) in Artificial Intelligence | Expected March 2027

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Generative AI, Computer Vision, Data Mining

Godavari Institute of Engineering and Technology, India — Bachelor of Technology (B.Tech.) in Civil Engineering | 2018–2021